EX-AI-1.py - C:/Users/user/Downloads/EX-AI-1.py (3.13.1)

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```
from collections import deque

graph = {
    'A': ['B', 'C'],
    'B': ['D', 'E'],
    'C': ['F', 'G'],
    'D': [],
    'E': [],
    'F': [],
    'G': []
}

def bfs(start, goal):
    queue = deque([(start, [start])])
    visited = set()

    print("BREADTH FIRST SEARCH")
    print("Starting Node:", start)
    print("Goal Node:", goal)

    while queue:
        node, path = queue.popleft()

        if node in visited:
            continue

        visited.add(node)
        print("Visiting Node:", node)
        print("Current Path:", path)

        if node == goal:
            print("Goal Reached!")
            print("Shortest Path:", " -> ".join(path))
            print("Total Steps:", len(path) - 1)
            print("Visited Nodes:", visited)
            return

        for neighbor in graph[node]:
            if neighbor not in visited:
                queue.append((neighbor, path + [neighbor]))
```

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IDLE Shell 3.13.1

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```
Python 3.13.1 (tags/v3.13.1:0671451, Dec  3 2024, 19:06:28) [MSC v.1942
64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
========== RESTART: C:/Users/user/Downloads/EX-AI-1.py ==========
========
BREADTH FIRST SEARCH
Starting Node: A
Goal Node: G
Visiting Node: A
Current Path: ['A']
Visiting Node: B
Current Path: ['A', 'B']
Visiting Node: C
Current Path: ['A', 'C']
Visiting Node: D
Current Path: ['A', 'B', 'D']
Visiting Node: E
Current Path: ['A', 'B', 'E']
Visiting Node: F
Current Path: ['A', 'C', 'F']
Visiting Node: G
Current Path: ['A', 'C', 'G']
Goal Reached!
Shortest Path: A -> C -> G
Total Steps: 2
Visited Nodes: {'B', 'E', 'C', 'D', 'A', 'G', 'F'}
>>>
```

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